Aslihan Karatas-Fredenburg, akaratas@uic.edu uicbeilab.github.io

Education

University of Michigan
Postdoctoral Research Fellow, Aug 2014–Aug 2015
Civil and Environmental Engineering
Supervisor Name: Carol Menassa

University of Illinois Urbana-Champaign
Doctor of Philosophy
Civil and Environmental Engineering, May 2014
Dissertation Title: Optimizing the Sustainability of Single-Family Housing Units
Advisor Name: Khaled El-Rayes

University of Florida
Master of Science
Civil and Coastal Engineering, May 2009
Thesis Title: Survey of Supply Chain Management as Perceived by the US Construction Industry
Advisor Name: Ralph Ellis

Bogazici University
Bachelor of Science
Civil Engineering, June 2007

Professional Appointments

University of Illinois Chicago Fall 2023–Present
Assistant Professor, Department of Civil, Materials, and Environmental Engineering
Principal Investigator of Built Environment and Infrastructure Laboratory (BEI Lab)

University of Illinois Chicago Fall 2021–Spring 2023
Clinical Assistant Professor, Department of Civil, Materials, and Environmental Eng.
Principal Investigator of Built Environment and Infrastructure Laboratory (BEI Lab)

Lawrence Technological University Fall 2015–Spring 2021
Assistant Professor, Department of Civil and Architectural Engineering

Walbridge Construction Company May 2019–Feb 2021
Project Controls Engineer at Scheduling Group
Research Analysts at Innovation Improvement Leadership Group

Warsaw University of Technology June 2006–August 2006
Undergraduate Research Assistant, Department of Civil Engineering

Research Grants (Pending)

Robert Wood Johnson Foundation (RWJF) Under Review
Title: LOI: When Silence Becomes Data: Community-Governed Alternative to Survey-Only Housing Health Research
Role: Principal Investigator
Co-PI: Michael Calias
Budget: $199,639

Housing and Urban Development (HUD) Under Review
Title: Pre-Proposal: Quantifying Cost, Compliance and Affordability Impacts of Energy-

Code Requirements in Affordable Home Construction
Role: Principal Investigator
Budget: $290,000

National Science Foundation (NSF) Under Review
Title: Physics-Informed Hybrid Quantum-Classical Reduced-Order Modeling of Environmental Risks from Air Leakage and Moisture in Building Wall Systems Under Future Weather Conditions
Role: Principal Investigator
Co-PIs: Ahmet Enis Cetin
Budget: $775,000

National Science Foundation (NSF) Under Review
Title: CPS -CIR: AI-Powered Thermal Imaging for Cyber-Physical Measurement of Window Air Leakage in Aging Housing Stock
Role: Principal Investigator
Co-PIs: Lucio Soibelman, Ahmet Enis Cetin, Lauryn Spearing, Yu Hou
Budget: $826,639

National Science Foundation (NSF) Under Review
Title: CAREER: Building Aging Dynamics Modeling for Energy Resilience: Integrating Building Degradation, Maintenance Behavior, and Climate Adaptation
Role: Principal Investigator
Budget: $599,999

National Science Foundation (NSF) Under Review
Title: Forecasting Long Term Impact of Flash Floods on Timber Walls using Explainable Machine Learning Models
Role: Principal Investigator
Co-PIs: Sourav Medya, Didem Ozevin
Budget: $499,999

Tissue Bank Asbestos Research Charitable Trust (TBAR) Under Review
Title: Mechanical Insulation Energy Savings and Efficiency ROI Scenarios for Owner and Contractor Decision-Making
Role: Principal Investigator
Co-PIs: Sogand Hasanzadeh, Behzad Esmaeili
Budget: $160,996

Tissue Bank Asbestos Research Charitable Trust (TBAR) Under Review
Title: Immersive Training Content for Mechanical Insulation â 360Â° Scenario-Based Learning
Role: Principal Investigator
Co-PIs: Sogand Hasanzadeh, Behzad Esmaeili
Budget: $200,000

STO Corp. Under Review
Title: Digital Twin Platform for Virtual Testing of STO Corp Wall Systems and Materials
Role: Principal Investigator
Budget: $463,240

STO Corp. Under Review
Title: Full Life Cycle Analysis Platform for STO Corp Wall Systems and Materials
Role: Principal Investigator
Budget: $185,443

STO Corp. Under Review
Title: Comparative Analysis of Theoretical and Experimental Thermal Performances and Water/Air Resistance of Wall Panel Joints
Role: Principal Investigator
Budget: $221,321

Research Grants (Ongoing)

U.S. Department of Housing and Urban Development (HUD) Ongoing
Title: Advancing Urban Resilience: The MoldMap Development for Predictive Risk Assessment in Low-Income Neighborhoods
Role: Principal Investigator
Co-PIs: Michael Cailas, Andrew Foell, Apostolis Samabanis
Budget: $998,764 — Duration: September 2024–September 2027

Tisue Bank Asbestos Research Charitable Trust (TBAR) Ongoing
Title: Synthesizing Current Challenges in Aging Infrastructure for Mechanical Heat and Frost Insulation Workforce
Role: Principal Investigator
Co-PIs: Sogand Hasanzadeh, Behzad Esmaeili
Budget: $61,996 — Duration: April 2024–December 2026

National Science Foundation (NSF) Ongoing
Title: Removing the Disparity in Success-Related Outcomes Between Academically Talented Low-Income Engineering Students and Other Engineering Students
Role: Senior Personnel
PI/Co-PIs: Houshang Darabi, Pete Nelson, Betul Bilgin, Shanon Reckinger, Renata A. Alonso, Miiri Kotche
Project Budget: $2,500,000 — Duration: June 2024–June 2027

SOL Projex Ongoing
Title: Environmental Performance Testing of SOL Projexâs Bio-Based Albizia–Lime Composite for Building Applications, Phase I
Role: Principal Investigator (Sole PI)
Budget: $40,262 — Duration: March 2026–December 2026

Research Grants (Completed)

USG Inc. Completed
Title: Water Penetration Resistance Test on USG Panels
Role: Principal Investigator (Sole PI)
Budget: $33,535 — Duration: September 2025–March 2026

US Department of Energy/Oak Ridge National Lab (DOE ORNL) Completed
Title: Finding the Impact of Window Replacement on Householdsâ Air Leakage
Role: Principal Investigator (Sole PI)
Budget: $72,950 — Duration: August 2023–August 2026

STO Corp. Completed
Title: Comparing Thermal Performances of Joints on Prefab Walls and Precast Walls
Role: Principal Investigator (Sole PI)
Budget: $222,550 — Duration: August 2023–February 2026

National Science Foundation (NSF) Completed
Title: CIVIC-PG Track B: Community-Driven, AI-Powered Thermal Imaging for Accessible Window Air Infiltration and Leakage Measurement
Role: Principal Investigator
Co-PIs: Lucio Soibelman, Ahmet Enis Cetin, Lauryn Spearing, Yu Hou
Project Budget: $75,000 — Duration: September 2024–March 2025

UIC Chancellor's Translational Research Initiative Award Completed
Title: Window Diagnostic Tool for Occupants' Safety and Indoor Environmental Quality in Commercial Buildings
Role: Principal Investigator
Co-PI: Enis Cetin
Budget: $25,000 — Duration: August 2024–August 2025

STO Corp. Completed
Title: Comparing Actual and Theoretical Environmental Performances of Prefab Wall Assemblies
Role: Principal Investigator (Sole PI)
Budget: $212,000 — Duration: April 2022–December 2023

UIC COVID Relief Funding Completed
Title: Investigating Hygrothermal Performance in Building Walls Engineered for Extreme Cold Climate Environments
Role: Principal Investigator (Sole PI)
Budget: $15,000 — Duration: August 2023–August 2024

UIC Teaching Sustainability Initiative Completed
Title: Developing a Module for CME 481 Sustainable Construction Course
Amount: $1,000

2023 ASCE Remote ExCEEd Teaching Workshop Scholarship Completed
Amount: $500 — Duration: July 18–27, 2023

Center for Construction Research and Training (CPWR) Completed
Title: Improving the Assessment of Noise Exposure and Warning Signal Audibility on Construction Sites
Role: Co-Principal Investigator
PI/Co-PIs: Nikolina Samardzic , Behzad Esmaeili
Total Budget: $25,000 — Co-PI Budget: $5,000
Duration: August 2021–February 2023

Centrepolis Accelerator Completed
Title: IEQ-EP: Indoor Environmental Quality and Energy Performance System for Buildings
Role: Principal Investigator
Budget: $2,000 — Duration: January 2021–April 2021

Kern Family Foundation (KEEN) Completed
Title: Course Module Development Grant
Amount: $3,000 — Year: 2017

Kern Family Foundation (KEEN) Completed
Title: PBL/ACL Development Grant
Amount: $2,000 — Year: 2016

LTU Presidential Undergraduate Research Award Completed
Title: Identifying Effective Occupancy Interventions to Reduce Energy in Commercial Buildings
Role: Principal Investigator
Budget: $1,000 — Year: 2016

Journal Publications

[1] Khaleghi, H., Safarpour, H., Smith, B., Imburgia, K., Atasoy, L., Spearing, L. A., & **Karatas, A.** (2026). Living with leaky windows in aged homes: A community-driven analysis of lived experiences, coping strategies, and support needs in Chicago. *Building and Environment*, *301*, 114816.

[2] Palani, H., Bhandari, M., & **Karatas, A.** (2026). Experimental and data-driven characterization of window-induced air leakage in residential buildings. *Results in Engineering, 10926*.

[3] Khaleghi, H., Ozevin, D., & **Karatas, A.** (2025). Experimental framework for nondestructive assessment of argon gas content in insulated glass units. *Journal of Architectural Engineering, 31*(4), 04025037.

[4] Khaleghi, H., Xu, C., Ozevin, D., & **Karatas, A.** (2025). Quantifying argon concentration within insulating glass units using low-frequency ultrasonic technique. *Measurement, 252*.

[5] Palani, H., Salonvaara, M., & **Karatas, A.** (2025). Digital twin platform for evaluating thermal bridging effects on aggregate thermal performance of prefabricated wall panel systems. *Applied Thermal Engineering, 270*, 126216.

[6] Palani, H., & **Karatas, A.** (2025). A state-of-the-art review of the implementation of continuous insulation in US buildings based on energy codes. *Journal of Structural Design and Construction Practice, 30*(2), 03125002.

[7] Khaleghi, H., & **Karatas, A.** (2024). Assessing the dynamic thermal performance of prefabricated wall panels in extreme hot weather conditions. *Journal of Building Engineering*.

[8] Palani, H., & **Karatas, A.** (2024). Investigating the disparities between experimental and computational analyses of thermal performance in prefabricated wall panels. *Applied Thermal Engineering*.

[9] Samardzic, N., Best, V., Hammond, C., **Karatas, A.**, Esmaeili, B., & Sarsam, H. (2024). Construction site noise exposure assessment using binaural measurements and analysis. *Safety, 10*(4), 92.

[10] Palani, H., Acosta, J. G., **Karatas, A.**, & Derrible, S. (2023). The role of socio-demographic and economic characteristics on energy-related occupant behavior. *Jour-*

nal of Building Engineering.

[11] Palani, H., & **Karatas, A.** (2023). Innovative environmental chamber construction for accurate thermal performance evaluation of building envelopes in varied climates. *Buildings, 13*(5), 1259.

[12] Palani, H., Khaleghi, H., Salehi, P., & **Karatas, A.** (2023). Assessing hygrothermal performance in building walls engineered for extreme cold climate environments. *Sustainability, 15*(24), 16597.

[14] Acosta, J. G., Palani, H., Movahedi, A., **Karatas, A.**, & Derrible, S. (2023). Residential electricity consumption patterns and their relationship to commute times by mode. *Findings.*

[15] Rener, A., **Karatas, A.**, & Videan, B. (2022). Innovative design and execution model for improving productivity of interior prefabricated commercial wall assemblies. *Buildings, 13*(1), 68.

[16] Palani, H., & **Karatas, A.** (2022). Holistic approach for reducing occupantsâ energy consumption in hotel buildings. *Journal of Cleaner Production, 365*, 132679.

[17] Palani, H., & **Karatas, A.** (2021). Identifying energy-use behavior and energy-use profiles of hotel guests. *Applied Sciences, 11*(13), 6093.

[18] Li, D., Menassa, C. C., & **Karatas, A.** (2017). Energy use behaviors in buildings: Towards an integrated conceptual framework. *Energy Research & Social Science, 23*, 97â112.

[19] **Karatas, A.**, & El-Rayes, K. (2016). Optimal tradeoffs between housing cost and environmental performance. *Journal of Architectural Engineering.*

[20] **Karatas, A.**, Stoiko, A., & Menassa, C. C. (2016). Framework for selecting occupancy-focused energy interventions in buildings. *Building Research & Information, 44*(5â6), 535â551.

[21] **Karatas, A.**, & El-Rayes, K. (2015). Evaluating the performance of sustainable development in urban neighborhoods based on the feedback of multiple stakeholders. *Sustainable Cities and Society, 14*, 374â382.

[22] **Karatas, A.**, & El-Rayes, K. (2015). Optimizing tradeoffs among housing sustainability objectives. *Automation in Construction, 53*, 83-94.

[23] **Karatas, A.**, & El-Rayes, K. (2015). Parallel computing framework for optimizing environmental and economic performances of housing units. *Journal of Computing in Civil Engineering.*

[24] **Karatas, A.**, & El-Rayes, K. (2014). Optimal trade-offs between social quality of life and life-cycle cost in housing units. *Journal of Construction Engineering and Management, 140*(12).

Book Chapters

[1] Alsharif, S., & **Karatas, A.** (2022). Data-driven approach for improving schedule and cost estimation of nuclear power plant projects. In *Nuclear power plants: Recent progress and future directions.* Nova Science Publishers. ISBN: 978-1-68507-686-3.

[2] **Karatas, A.**, Stoiko, A., & Menassa, C. C. (2019). Framework for selecting occupancy-focused energy interventions in buildings. In *Building governance and climate change* (pp. 535â551).

[3] **Karatas, A.**, Stoiko, A., & Menassa, C. C. (2017). A framework to achieve large-scale energy savings for building stocks through targeted occupancy interventions: Foundations, principles, and applications. In *Smart cities: Foundations, principles, and applications*. John Wiley & Sons. https://doi.org/10.1002/9781119226444.ch17

Peer-Reviewed Conference Publications

[1] Salehi, P., & **Karatas, A.** (2026). Year-by-year dynamic building energy assessment with future climate projections capturing extremes and fluctuations. In *Proceedings of SimBuild Conference 2026* (Vol. 12, pp. 856â864). IBPSA-USA. https://doi.org/10.26868/30680611.2026.1376

[2] Salehi, P., Esmaeili, B., Hasanzadeh, S., & **Karatas, A.** (2025). Mold challenges in building mechanical systems faced by mechanical insulators. In *International Conference on Moisture in Buildings* (pp. 113â125). Cham, Switzerland: Springer Nature.

[3] Salehi, P., Pipim, C., & **Karatas, A.** (2025). Integrating household characteristics, mold inspection, and remediation data to develop localized mold-risk profiles for residential buildings. In *International Conference on Moisture in Buildings* (pp. 553â567). Cham, Switzerland: Springer Nature.

[4] Rong, S., Hamdan, E., Khaleghi, H., **Karatas, A.**, & Cetin, A. E. (2025). Air leak detection using Sobel-enhanced YOLO algorithm from infrared images. In *IEEE International Symposium on Circuits and Systems (ISCAS)* (pp. 1â5).

[5] Khaleghi, H., Rong, S., Hamdan, E., Hou, Y., Soibelman, L., Cetin, A. E., & **Karatas, A.** (2025). Towards an automated method for detecting and quantifying window air leakage. In *ASCE International Conference on Computing in Civil Engineering Computational and Intelligent Technologies* (pp. 384-390).

[6] Salehi, P., & **Karatas, A.** (2025). Towards climate change adaptive whole-building energy simulation models. In *ASHRAE Winter Conference*, Orlando, FL.

[7] Khaleghi, H., Salehi, P., Ozevin, D., & **Karatas, A**. (2025). Nondestructive Assessment for Detection of Low-E Coatings on Insulating Glass Units. In *CIB Conferences* (Vol. 1, No. 1, p. 233).

[8] Palani, H., Salonvaara, M., & **Karatas, A.** (2025).. Improving Thermal Performance of Prefabricated Wall Systems by Minimizing Thermal Bridging of Steel Brackets and Connectors with Minimal Investment Cost. *ASHRAE Transactions*, *131*, 98-105.

[9] Khaleghi, H., Salehi, P., Ozevin, D., & **Karatas, A.** (2025). Nondestructive Assessment for Detection of Low-E Coatings on Insulating Glass Units. In *CIB Conferences* (Vol. 1, No. 1, p. 233).

[10] Palani, H., Vo, T., Bhandari, M., & **Karatas, A**. (2025). A Review of the State-of-the-Art of Rainscreen Cladding Performance in Residential Building Walls. In *CIB Conferences* (Vol. 1, No. 1, p. 232).

[11] Palani, H., **Karatas, A.**, & Taylor, T. (2024). Cost-effective hot box apparatus to measure thermal performance of building envelopes. In *Construction Research Congress 2024* (pp. 739â747).

[12] Khaleghi, H., & **Karatas, A.** (2024). An experimental framework to measure dynamic thermal resilience of wall panels under extreme weather conditions. In *Construction Research Congress 2024* (pp. 327â335).

[13] Luna, H., Palani, H., & **Karatas, A.** (2023). Cost-effective, scalable, and user-friendly temperature control system for measuring thermal performance of building envelope systems in environmental chambers. In *International Conference on Computing in Civil Engineering* (pp. 815â823).

[14] Khaleghi, H., & **Karatas, A.** (2023). Arctic architectures: Unleashing energy efficiency and resilience in extreme cold regions. In *International Structural Engineering & Construction, 10*(1).

[15] Palani, H., Acosta, J., Derrible, S., & **Karatas, A.** (2023). Predicting energy use intensity of US hotel buildings using CBECS microdata. In *International Structural Engineering & Construction, 10*, 1â6.

[16] Rener, A., **Karatas, A.**, & Cole, M. (2022). Innovative model for forecasting trailer usage for prefabricated exterior wall panels. In *ISARC 2022*, Bogota, Colombia.

[17] Palani, H., Acosta, J., **Karatas, A.**, & Derrible, S. (2022). Understanding the interrelationship between energy-use behavior and socio-demographic profile. In *ITISE-2022*, Gran Canaria, Spain.

[18] Palani, H., & **Karatas, A.** (2022). Integrated energy-use model to identify energy-use profile of hotel guests. In *Computing in Civil Engineering* (pp. 688â695).

[19] Palani, H., & **Karatas, A.** (2021). Identifying energy use behaviors of hotel guests. In *CSCE 2021*.

[20] **Karatas, A.**, & Al-Sharif, S. (2020). Schedule and cost forecasting model for nuclear power plant projects. In *Construction Research Congress 2020* (pp. 721â729).

[21] Palani, H., & **Karatas, A.** (2020). Reducing occupantsâ energy consumption in hotel buildings. In *Building Performance Analysis Conference and SimBuild*.

[22] Albayati, A., & **Karatas, A.** (2019). Fall prevention supplementary devices for bridge construction workers: A life cycle cost analysis. In *Canadian Society for Civil Engineering International Construction Specialty Conference*, Montreal, Canada.

[23] Staple, N., & **Karatas, A.** (2019). A study of the U.S. roofing industry and its workforce. In *Canadian Society for Civil Engineering International Construction Specialty Conference*, Montreal, Canada.

[24] Alwerfalli, D., **Karatas, A.**, & Alshammari, M. (2017). Application of quality management systems (QMS) in construction industry. In *International Conference on Industrial Engineering and Operations Management*.

[25] Alsharif, S., & **Karatas, A.** (2016). A framework for identifying causal factors of delay in nuclear power plant projects. *Procedia Engineering, 145*, 1486â1492.

[26] **Karatas, A.**, Menassa, C. C., & Stoiko, A. (2015). A framework for delivering targeted occupancy interventions to reduce energy usage in buildings. *Procedia Engineering, 118*.

[27] **Karatas, A.**, & El-Rayes, K. (2014). Optimal configurations of design and construction decisions for maximizing social quality-of-life for housing residents. In *Construction Research Congress 2014* (pp. 436â443).

[28] **Karatas, A.**, & El-Rayes, K. (2014). Maximizing occupantsâ comfort in affordable housing units. In *Computing in Civil and Building Engineering* (pp. 2040â2046).

[29] **Karatas, A.**, & El-Rayes, K. (2012). Evaluating and quantifying sustainable development in urban neighborhoods. In *ICSDEC 2012* (pp. 214â221).

Professional Registrations and Memberships

- LEED Accredited Professional (AP BD+C), Credential ID: #10773918-AP-BD+C
- Certificate for Excellence in Sustainable Management and Technology, University of Illinois Urbana–Champaign, College of Business
- National Association of Women in Construction (NAWIC)
- U.S. Green Building Council (USGBC)
- American Society of Civil Engineers (ASCE), Associate Member
- American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)

Professional Service and Leadership

International and National Service

- *Invited Panelist*, Thought Leaders Policy Conference and Black Caucus Foundation Michigan Briefing, White House, Washington, DC — September 2016
- *Award Recipient*, National Association of Women in Construction (NAWIC) Foremost Leadership Award — 2019
- *Faculty Team Member*, Department of Energy (DOE) JUMP into STEM Program — 2022–2024
- *Editor*, *Proceedings of International Structural Engineering and Construction*, Vol. 10(1) — 2023–Present
- *Judge*, Department of Energy (DOE) JUMP into STEM Program — 2023–2024
- *Working Commission Member*, CIB W040 – Heat and Moisture Transfer in Buildings — 2025–Present
- *Panel Reviewer*, National Science Foundation (NSF) — Ongoing
- *Journal and Book Reviewer*, Springer Nature SNAPP; *Journal of Infrastructure Systems*; *Buildings*; *Energy & Buildings* — Ongoing

Conference Leadership

- *Technical Session Chair*, International Conference on Sustainable Design Engineering and Construction, Tempe, AZ — May 2016
- *Technical Session Chair*, CSCE Sustainable Design Engineering and Construction, Montreal, Canada — June 2019
- *Technical Session Chair*, International Conference on Time Series and Forecasting, Gran Canaria, Spain — June 2022
- *Technical Session Chair*, ISARC, Bogota, Colombia — July 2022
- *Conference Chair*, ISEC-12, International Structural Engineering and Construction Conference — August 2023

- *Technical Session Chair*, Construction Research Congress, IA — March 2024
- *Technical Session Chair*, ASHRAE Winter Conference, FL — February 2025
- *Technical Session Chair*, Construction Research Congress, TX — March 2026

University and Departmental Service

- *Subcommittee Member*, Office Furniture Subcommittee, University of Illinois Chicago — 2023–2024
- *Jury Member*, Chris and Susan Burke UIC Graduate Student Poster Competition — 2023; 2025
- *Mentor*, NSF S-STEM Undergraduate Student, UIC Department of Civil, Materials, and Environmental Engineering — 2024–Present
- *Committee Member*, UIC CME Graduate Research and Teaching Awards — April 2025
- *Committee Member*, UIC Chancellor's Committee on Sustainability and Energy — 2025–Present
- *Committee Member*, UIC CECHE Air Pollution, Extreme Heat, and Public Health Workgroup — 2026–Present

Prior Institutional Leadership

- *President*, Turkish Student Association, University of Florida — 2008–2009
- *Academic Advisor and Advisory Board Member*, The BUILD Initiative — 2015–2021
- *Faculty Senate Member*, Lawrence Technological University — 2017–2020
- *Campus Affairs Committee Chair*, Lawrence Technological University — 2018–2019

Courses Taught

University of Illinois Chicago — Fall 2021–Present

- CME 112 – Evolution of Infrastructure
- CME 201 – Statics
- CME 481 – Decision Making and Risk Management in Construction
- CME 488 – Sustainable Construction
- CME 587 – Construction Estimating and Scheduling
- CME 594 – Pre-Construction and Contract Administration

Student Advising and Mentoring

Doctoral Students – Chair/Advisor

- *Samer Al-Sharif*, Ph.D., Lawrence Technological University. Dissertation/project: *Schedule and Cost Forecasting Model for Nuclear Power Plant Projects.* Completed. — Spring 2017
- *Andrew Rener*, Ph.D., Lawrence Technological University. Dissertation: *Innovative Methodologies for Enhancing Productivity, Supply Chain, and Sustainability in Industrialized Construction.* Completed. — Fall 2022

- *Hevar Palani*, Ph.D., Civil, Materials, and Environmental Engineering, UIC. Dissertation: *Integrative Assessment Methodologies for Sustainable Building Envelope Systems in Climate Adaptation*. Completed. — Spring 2025
- *Hamed Khaleghi*, Ph.D. student, Civil, Materials, and Environmental Engineering, UIC. Dissertation: *Novel Diagnostic Methods and Performance Evaluation Technique for Resilient Window Systems*. Current. — Fall 2022–Present
- *Parisa Salehi*, Ph.D. student, Civil, Materials, and Environmental Engineering, UIC. Dissertation: *Developing Sustainable Building Practices in Response to Future Weather Conditions*. Current. — Fall 2023–Present
- *Nishat Tasnim*, Ph.D. student, Civil, Materials, and Environmental Engineering, UIC. Dissertation: *Measuring Air Tightness of Building Envelope Systems with AI Assisted Analysis*. Current. — Fall 2025–Present
- *Elhassan Reda Ismail*, Ph.D. student, Civil, Materials, and Environmental Engineering, UIC. Dissertation: *Advancing Urban Resilience with MoldMap Development for Predictive Risk Assessment*. Current. — Fall 2025–Present

Graduate Committees

- *Fatemeh Allahakbari*, Ph.D., Committee Member — Spring 2023–Present
- *Shuaiang Rong*, Ph.D., Committee Member — Fall 2023–Spring 2026
- *Lubna Shibly Mokatren*, Ph.D., Committee Member — Fall 2020–Fall 2025

Master's Students

- *Abdulkadir Ismail*, MSc, Independent Research Study Advisor — Spring 2022

Undergraduate Research Students

- *Karolina Dorula*, UIC CURA Awardee — Spring 2022
- *Jildardo Luna*, UIC CURA Awardee — Spring 2022–2023
- *Michael Nwaigbo*, Honors College Fellow — Spring 2023
- *Hoang Bao Nghi Bien*, Independent Research — Fall 2023–2024
- *Chris Amankwah*, UIC GPIP Intern — Summer 2024–2025
- *Shaan Mehta*, UIC GPIP Intern — Summer 2024
- *Thanh Vo*, Honors College Fellow — Fall 2024–Spring 2025
- *Bennett Smith*, Honors College Fellow — Fall 2024–Fall 2025
- *Jamie Frost*, Independent Research — Fall 2024
- *Kelly Myriam*, UIC CURA Awardee — Spring 2025–2026
- *Gabriel Cajiao*, Independent Research — Spring 2026
- *Emily Fallaw*, Honors College Fellow Current. — Spring 2026–Present

Visiting Scholars / International Research Interns

- *Alexandre Pebayle*, CESI Engineering School, France, Visiting Research Intern. June–October 2024
- *Louis Bonnin*, CESI Engineering School, France, Visiting Research Intern. June–October 2025
- *Tom Routier*, CESI Engineering School, France, Visiting Research Intern. June–October 2025

Postdoctoral Research Supervision

- *Haluk Isik*, Postdoctoral research supervision and mentorship. Current. September 2026–Present